Modelling Moon Phases · TA Brief

Learning goal: I can test a 3D model of the Moon's orbit to record all eight phases and evaluate the model.

Watch (staff preview first)

Short silent teacher demonstration clip for this lesson's model and practical sequence · preview before class use.

Adult Support Routine

No cold calling; nobody is asked to read difficult text aloud.

Give time to start -> point to the relevant place -> rephrase -> offer a small prompt or word bank -> reduce to a limited choice.

Beside the You do in the pupil book, mark I (independent), S (supported) or C (choice).

Must-Know Words

waxing crescent: the phase after New Moon when a thin sliver of light grows on the right

waxing gibbous: the phase before Full Moon when more than half of the face is lit and growing

waning gibbous: the phase after Full Moon when more than half is lit and shrinking

waning crescent: the final thin sliver of light before the cycle returns to New Moon

Key Answers

A. Without tilting the orbit slightly, the observer's head blocks the torch at Position 5 and creates a lunar eclipse instead of a Full Moon.

A. Between First Quarter (Day 7) and Full Moon (Day 14), the lit share continues growing through Waxing Gibbous.

1. Torch = Sun, Observer = Earth, Ball = Moon. 2. New Moon -> First Quarter -> Full Moon -> Third Quarter (with Waxing/Waning Crescent and Gibbous in between). 3. Strength: shows in 3D that half the Moon is always lit while our viewing angle changes. Limitation: sizes and distances are not to scale, and Earth's daily spin/orbit motion are simplified.